PHYSICS PRACTICAL MANUAL

(FIRST YEAR SECOND SEMESTER)

Elementary Laboratory

Department of Physics

University of Sri Jayewardenepura

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Experiment No	Experiment
01	Determination of Refractive Indices of Solids and Liquids using Different Optical Instruments
02	Determination of viscosity of an unknown liquid using Stoke’s method
03	Determination of End Corrections, Resistance of an Unknown Resister, Resistivity of a Conducting Wire using the Meter Bridge
04	Analyzing Digital Circuits
05	Deflection Magnetometer Experiment
06	Comparison of Magnetic Moments / Deflecting Magnetometer
07	Voltage Regulation
08	Operational Amplifier(OP-AMP)
09	Bi-filler suspension
10	Characteristics of Diodes and Semi conductor properties (Determination of Plank’s and Boltzmann constants)

Experiment No: 01

Determination of Refractive Indices of Solids &Liquids using Different Optical Instruments.

1.1. Determination of refractive indices of glass & water using a glass prism (Critical Angle method)

Theory:

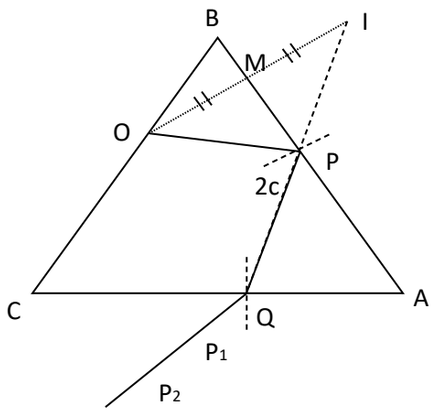


Figure 1.1

If μ_g is the refractive index of glass

$$\mu_g = \frac{1}{\text{Sin C}} \quad \text{----------} \quad 1.1$$

Apparatus

Prism, Microscopic slide, 3 pins

Exercise 01

1. Place the Glass prism on the center of the sheet of paper.
2. By marking the edges A, B, C of the prism on the paper, trace the exact shape of the prism.
3. Now fix a pin, O, close to the center of the face BC of the prism such that the pin touches face BC
4. Observe the image of the pin O through face AC of the prism. Now move your viewing position from A to C. As you move from A to C at a particular position the image of O through face AC will disappear.
5. At the above position fix two pins P_1 and P_2 some distance apart, along the line joining the image I and your eye.
6. Remove the prism and draw a line passing through P_1&P_2
7. Draw a perpendicular from point O to line AB and extended it up to I such that OM=MI. connect QI.
8. Mark the point of intersection between line QI and AB as P. Now connect OP.
9. Measure the angle OPQ. Since OPQ=2c (c is critical angle) substitute the value of c in to equation 1.1 and obtain the refractive index of the prism material.
10. Place a microscope slide containing few drops of water in contact with face AB of the prism and repeat the experiment obtain a value for refractive index of water and find the error of n using error propagation formula.

1.2. Determination of refractive index of a liquid using concave mirror

Theory:

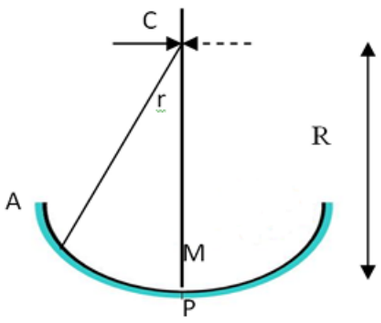


Figure 1.2

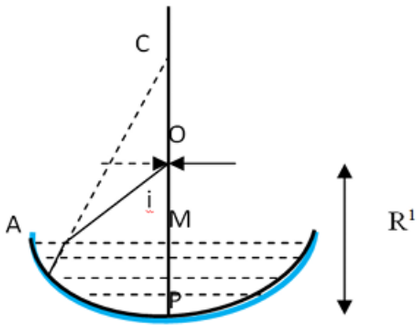


Figure 1.3

If μ_w is the refractive index of water,

$$\mu_w = \frac{Sin\ i}{Sin\ r} = \frac{PC}{PO} = \frac{R}{R^1} \text{ ---------- } 1.2$$

Exercise 02

1. Place a concave mirror on the table and adjust the position of a pin, situated above it, until the image and object coincide.
2. Measure the distance of the pin from the mirror (i.e. PC). This distance refers to the radius of curvature of the concave mirror.
3. Now put a thin layer of liquid in the mirror and again perform the experiment. Measure the distance to the new position of coincident from the mirror (i.e. PC).
4. From the equation 1.2 determine the refractive index of the liquid and find the error of μ_w using error propagation formula.

1.3. Determination of refractive index of a liquid using a convex lens

Theory:

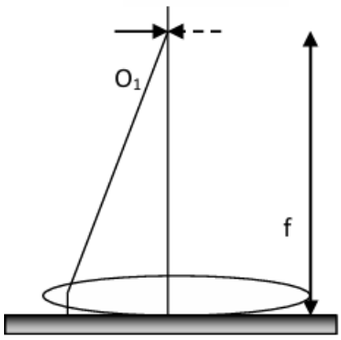


Figure 1.4

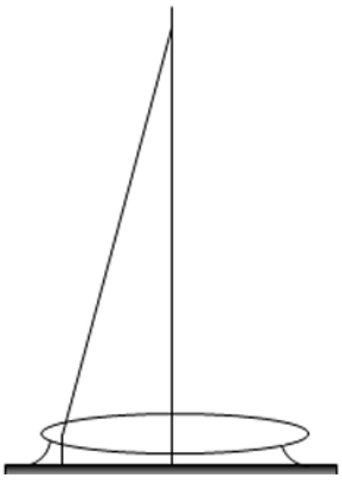


Figure 1.5

$1/F = 1/f_1 + 1/f_2$ for a liquid lens

$1/f_2 = (\mu_i - 1)(1/r_1 - 1/r_2)$ ---------- 1.3

F_2 : focal length of the liquid lens

μ_i : refractive index of the liquids

r_1, r_2 : radii of curvature of the liquid

Apparatus

Convex lens, Plane mirror, Stand, Pins

Exercise 03

1. Place a convex lens on the plane mirror and set 2 items on the table.
2. Adjust the position of the pin O_1 above the plus mirror systems until the image and object coincide.
3. Measure the distance O_1P which is equal to the focal length of the convex lens.
4. Now put a thin layer of given liquid the plane mirror and convex lens and obtain the new position of the coincidence for the image O_2 and object. Measure the distance O_2P, which is the focal length of the combined lens.
5. Measure the radius of the curvature of the lower surface of the convex lens using the Spherometer.
6. Determine the refractive index μ_i of the liquid using equation 1.3 and also find the error of μ_i using error propagation formula.

1.4. Determination of refractive index of a glass cube using a traveling microscope

Theory:



Figure 1.6 Figure 1.7 Figure 1.8

$$\mu_g = \frac{\text{True Depth}}{\text{Apparent Depth}} = \frac{T_3 - T_1}{T_3 - T_2} \quad \text{---------- 1.4}$$

Apparatus

Glass cube, travelling microscope, colored powder

Exercise 04

1. According to the figure 1.6 place some colored powder on the glass base of the traveling microscope and focus the microscope on to that. Take the corresponding reading T_1
2. Now place a glass slide on the top of the powder and move the microscope vertically until the colored powder is focused again. Take the corresponding reading T_2 according to figure 1.7.
3. Now put colored powder on top of the glass slide and move the microscope vertically so that the powder is focused again. Take the corresponding T_3 according to figure 1.8
4. From the equation 1.4 determine the refractive index of the glass slide and it's error using the error propagation formula.

Experiment No: 02

Determination of viscosity of an unknown liquid using Stoke's method

Introduction:

Viscosity is a property of a fluid by virtue of which an internal resistance comes into play when the liquid is in motion, and opposes the relative motion between its different layers. Thus, it is the resistance of a fluid to flow.

When liquid flows over flat surface, a backward viscous force (F) acts tangentially to every layer. This force depends upon the area of the layer, velocity of the layer, and the distance of the layer from the surface.

$$F \propto A\frac{dv}{dx}$$

$$F = \eta A\frac{dv}{dx}$$

Where η is the ***coefficient of viscosity*** of the liquid.

Theory:

Stoke's Law

Stoke's law was established by an English scientist Sir George G Stokes (1819-1903).

When a spherical body moves down through an infinite column of highly viscous liquid, it drags the layer of the liquid in contact with it. As a result, the body experiences a retarding force.

Then according to Stokes law, the viscous drag force,

$$F = 6\pi\eta r v$$

where, r - Radius of the spherical body

v - Velocity of the spherical body

It gives the relationship between retarding force and velocity. When viscous force plus buoyant force becomes equal to force due to gravity, the net force becomes zero. The sphere then descends with a constant terminal velocity (v_t) (Figure 2.1).

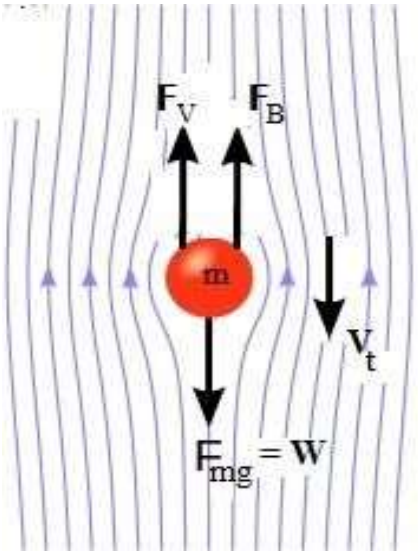


Figure 2.1

$$6\pi\eta r v_t + \frac{4}{3}\pi r^3 \sigma g = \frac{4}{3}\pi r^3 \rho g$$

Now,

$$6\pi\eta r v_t = \frac{4}{3}\pi r^3 (\rho - \sigma) g$$

$$\eta = \frac{2}{9}\frac{r^2(\rho - \sigma)g}{v_t}$$

$$v_t = \frac{2}{9}\frac{(\rho - \sigma)g}{\eta}.r^2$$

where, ρ - Density of the material of the spherical body

σ - Density of the liquid

Apparatus:

A long cylindrical lighted glass cylinder, Transparent viscous fluid, Meter ruler, Steal ball bearings, Micrometer screw gauge, Vernier caliper, Stop watch, Thread or rubber bands

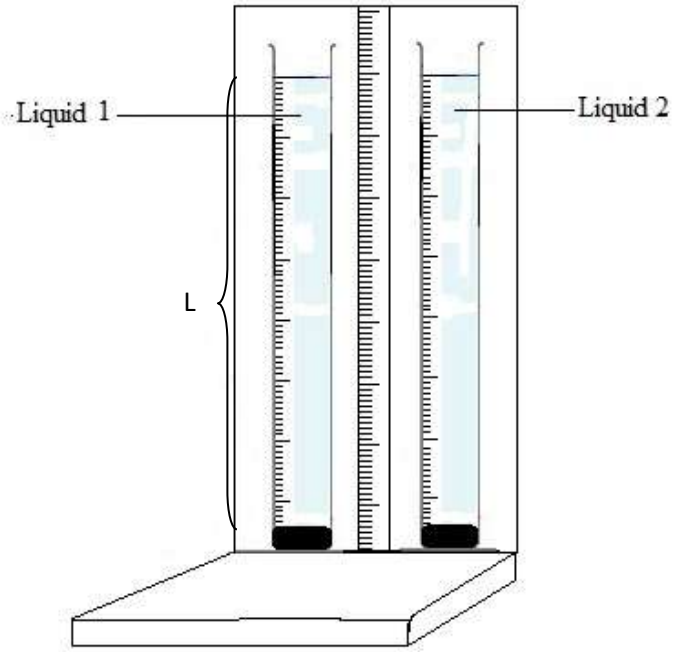


Figure 2.2

Exercise:

1. Determine the least count and the zero error of the screw gauge and vernier caliper.
2. You have provided 15 beads of three different sizes. Measure the mean diameter of the each size of the bead (Bead A group (three beads), Bead B group (three beads), and Bead C group (three beads)using the micrometer screw gauge.
3. Measure the inner diameter of the glass cylinder using the vernier caliper.

4. Take a bead and drop it gently in to the liquid 1. It falls down in the liquid with an accelerated velocity for some time and then moves with a uniform terminal velocity.
5. Using a thread, mark a reference point as A below the point at which the bead acquires terminal velocity. Mark another point as B below the point A using another thread.
6. Now take another bead and drop it gently into the liquid 1.
7. A stop watch is switched on when the bead crosses the point A, and switched off when it leaves the point B. Note the time taken by the bead to travel the distance AB.
8. Then the observed terminal velocity of the bead can be calculated using,

$$Observed\ Terminal\ Velocity\ (\boldsymbol{V}) = \frac{Distance\ AB}{\Delta t}$$

The corrected terminal velocity of the bead can be calculated using the relation,
(Using the empirical relationship of Newman and Searle)

$$V_t = V\left(1+\frac{2.4\,r}{R}\right)\left(1+\frac{3.3\,r}{h}\right)$$

Here; r = radius of the bead

R= inner radius of the cylinder

h= height of the AB

9. Note the time taken to travel the distance AB for each bead and fill the below table for each Bead size including all beads.

Bead A	Radius of the glass bead (r ±δr mm)	r^2 (mm^2)	Time t ±δt (s)	Observed terminal velocity V±δV (ms^{-1})	Corrected Terminal Velocity V_t ±δV_t (ms^{-1})
1					
2					
3					

Bead B	Radius of the glass bead (r ±δr mm)	r_2 (mm^2)	Time t ±δt (s)	Observed terminal velocity V±δV (ms^{-1})	Corrected Terminal Velocity V_t ±δV_t (ms^{-1})
1					
2					
3					

Bead C	Radius of the glass bead (r ±δr mm)	r_2 (mm^2)	Time t ±δt (s)	Observed terminal velocity V±δV (ms^{-1})	Corrected Terminal Velocity V_t ±δV_t (ms^{-1})
1					
2					
3					

10. Find the average coefficient of viscosity of the liquid 1 using

(i) Bead A group data, (ii) Bead B group data, and (iii) Bead C group data.

12. Follow the same procedure to find out the coefficient of viscosity of the two different liquids and calculate $\eta \pm \delta\eta$.
13. Use error theory in each step to figure out the corresponding error propagations and create appropriate tables to include error calculation for each bead data.

Experiment No: 03

Determination of End Corrections, Resistance of an Unknown Resistor, Resistivity of a Conducting Wire using the Meter Bridge

3.1. Determination of the end corrections

Theory:

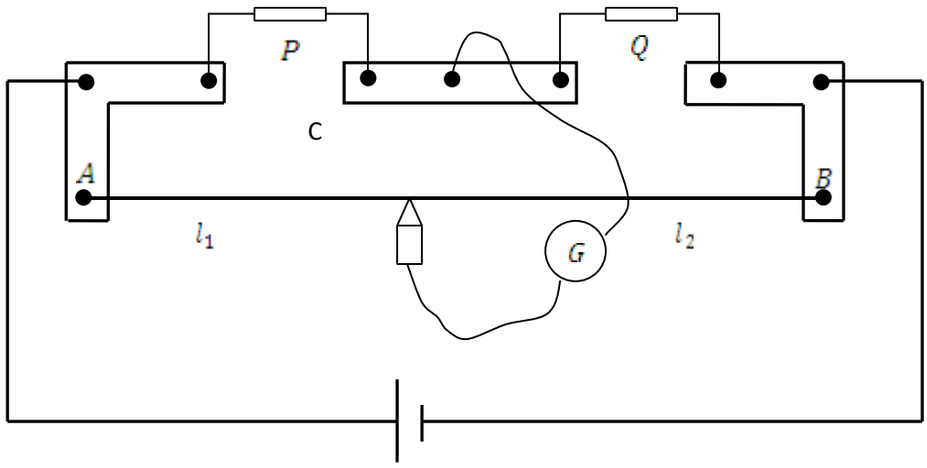


Figure 3.1

In the figure 4.1, when the deflection of *G*, a center zero galvanometer is zero for a perfect balance the resistance R_{AC} , R_{BC}, *P* &Q from a Wheatstone Bridge. That is,

$$\frac{P}{Q} = \frac{R_{AC}}{R_{BC}} \quad \text{---------- 3.1}$$

Where, R_{AC} = resistance of the length AC of the meter wire AC.

R_{BC} = resistance of the length BC of the meter Wire BC.

In the meter bridge, the resistance of end corrections and resistance of the metal connections are neglected.

Resulting errors due to the above are accounted for by introducing end corrections for the meter bridge. These end corrections are equivalent to resistances of length X_1 and X_2 at the two ends A and B of the meter wire.

Thus

$$R_{AC} = (l_1 + X_1)\,\rho \quad , \qquad l_1 = AC$$
$$R_{BC} = (l_2 + X_2)\,\rho \quad , \qquad l_2 = BC$$

ρ is the resistance per unit length of the wire AB.

Then from equation

$$\frac{P}{Q} = \frac{(l_1 + X_1)\rho}{(l_2 + X_2)\rho} = \frac{(l_1 + X_1)}{(l_2 + X_2)} \quad \text{---------- 3.2}$$

When P & Q are chosen such that *P*=*R* & *Q*= *nR* (*n* is an integer)

$$\frac{R}{nR} = \frac{(l_1 + X_1)}{(l_2 + X_2)}$$
$$l_2 + X_2 = n(l_1 + X_1)$$
$$100 - l_1 + X_2 = n(l_1 + X_1)$$
$$100 - l_1 - X_1 + X_1 + X_2 = n(l_1 + X_1)$$
$$100 + X_1 + X_2 = n(l_1 + X_1) + l_1 + X_1$$
$$l_1 = (100 + X_1 + X_2)\frac{1}{n+1} - X_1$$
$$y = mx + c$$

From the graph of $l_1\ vs\ \frac{1}{n+1}$ we can determine the end correction X_1 and X_2

Exercise 01

1. Connect a known resistor for P and connect the resistor box for Q.
2. Change the resistance of the resistor box such that it is always a multiple (n) of the known resistor value.
3. Obtain the corresponding balance length l_1, for each adjustment of the resistor box value.
4. Draw a suitable graph $l_1\ vs\ \frac{1}{n+1}$ hence determine the end correction X_1 and X_2.

3.2. Determination of the resistance of an unknown resistor

Theory:

In the figure 3.1 replace *P* by a resistor box and *Q* by the unknown resistor
Then from Eq^n:

$$P/Q = (l_1+X_1) / (l_2+X_2)$$

After measuring the balance length l_1,

$$P/Q = (l_1+X_1) / (100-l_1+X_2)$$

Finally we get,

$$\frac{1}{l_1 + X_1} = \frac{Q}{100 + X_1 + X_2}\frac{1}{P} + \frac{1}{100 + X_1 + X_2}$$

The graph $1/(l_1 + X_1)$ Vs. $1/P$ traces a straight line and its gradient is given by the resistance of the unknown resistor, *Q*

Exercise 02

1. Replace *P* and *Q* of the meter bridge by the resistor box and the unknown resistor, respectively.
2. Change the resistor value of the resistor box and measure the corresponding balance length on the meter bridge.
3. By plotting a suitable graph $1/(l_1+X_1)$ vs. $1/P$ calculates the resistance of the unknown resistor.

Note

Always make sure that the balance point comes fairly near to the center of the wire AB. Do not scrape the movable slider along the wire AB measure the deflection of the galvanometer by placing the movable slider from point to point.
Do not connect the meter bridge to a power supply for a long duration.

3.3. Determination of the resistivity of a given conducting wire

Theory:

The resistance R of a conducting wire of length l and cross area A is given by, $R=\rho l/A$
here ρ is the resistively of the conducting wire.
The graph of R vs. l is a straight line passing through the origin and its gradient gives the resistivity$\rho$of the conducting wire.

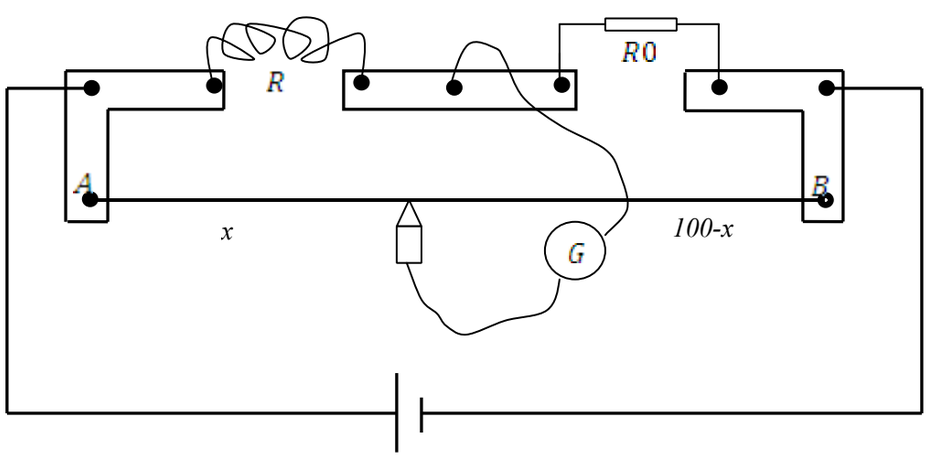


Figure 3.2

Exercise 03

1. Connect a known length (l) of a conducting wire to the left hand side gap on the meter bridge.
2. Similarly connect a resistor box to adjust the resistance (R_0) to the right hand side on the meter bridge.
3. Now by balancing the meter bridge obtain the corresponding length x. Then from, $R/R_0 = x/(100-x)$
4. Find value for R.
5. Now by changing the length (l) of the conducting wire attached to the left hand side gap on the meter bridge and find the corresponding resistance R by balancing the meter bridge.

Experiment No: 04

Analyzing a Digital Circuit.

Theory:

You are provided with a controlling IC (Integrated Circuit) of a video cassette player (The actual circuit is covered). The task of this IC is to turn on or turn off a servo motor connected to the output of the IC.

The block diagram of the actual controlling IC is given below.

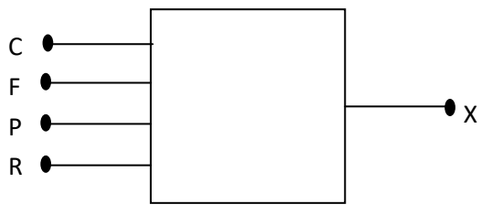


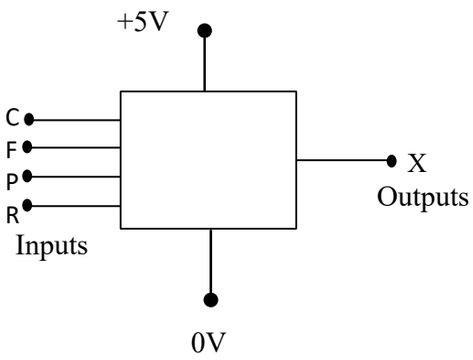
Figure 4.1

Video cassette player generates the four input signals as C, F, P and R, X is the output of the IC. The input signals are generated according to the following conditions.

C -- is designated as "Cassette is in the player signal"
F -- is designated as "Fast forward button depressed"
P -- is designated as "Play button pressed"
R -- is designated as "Rewind button pressed"

Exercise 01

1. Apply all the voltage combinations to inputs C, F, P and R and investigate the voltage of the output X and Construct the Truth Table given below.



Inputs				Output
C	**F**	**P**	**R**	**X**
0	0	0	0	
0	0	0	1	
0	0	1	0	
.	.	.	.	
.	.	.	.	

Figure 4.2

2. Now by studying the truth table write down the Boolean expression for output X.
3. Physically interpret each term in the output expression. (You can use the definitions given for the inputs C, F, P and R for this interpretation)

4.2. Designing a logic circuit for AC Water pump

A student wishes to design a digital circuit to turn on the switch of an electric water pump automatically, when the water tank is empty. In addition, the pump switches on only when the well is sufficiently filled with water. In an emergency the pump can be switched on by pressing a button.

The block diagram of the circuit, having three inputs and one output, is shown in the figure given below,

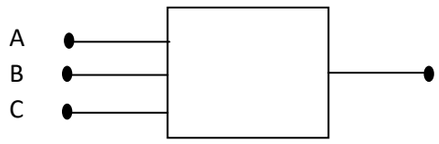


Figure 4.3

The following definitions are given for the three inputs A, B and C depending on the logic value 0 and 1.

When,

A = 0	the emergency button is not pressed
A = 1	the emergency button is pressed
B = 0	the water tank is empty
B = 1	the water tank is full
C = 0	the well is sufficiently filled with water
C = 1	the well isn't sufficiently filled with water

The circuit is to be designed such that,

When F = 1 the water pump is " ON ".
When F = 0 the water pump is " OFF ".

Exercise 02

1. Design the truth table for the above situation.
2. Write down the logic expression for output F in terms of the inputs A, B and C.
3. Draw a circuit diagram for the logic expression using basic logic gates and label A, B, C and F.
4. Construct the circuit using logic gates and verify the expression (derived in step 3 of Exercise 02) experimentally.

4.3 Relays

Theory:

A relay is an electro-magnetic switch, which is useful to use a low voltage circuit to switch on and off a light bulb (or anything else) connected to the 230 V mains supply.

Relays are very simple devices. There are four parts in every relay:

1. Electromagnet
2. Armature (that can be activated by the electromagnet)
3. Spring
4. Set of electrical contacts

The following figures show these four components in action:

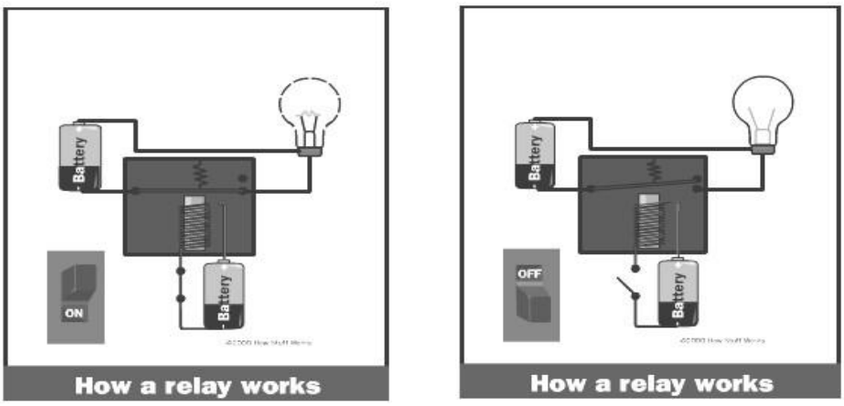
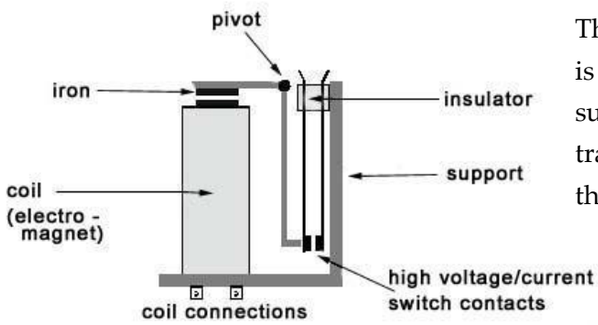


Figure 4.4

In this figure 4.4, you can see that a relay consists of two separate and completely independent circuits. The first is at the bottom and drives the electromagnet. In this circuit, a switch is controlling power to the electromagnet. When the switch is turned on, the electromagnet is activated, and it attracts the armature. The armature is acting as a switch in the second circuit. When the electromagnet is energized, the armature completes the second circuit and the light is on. When the electromagnet is not energized, the spring pulls the armature away and the circuit is not complete. In that case, the light is switched off.

The figure 4.5 shows a typical relay.



The current needed to operate the relay coil is higher than the amount that can be supplied by most chips (op. amps etc), so a transistor is usually needed, as shown in the figure 4.6.

Figure 4.5

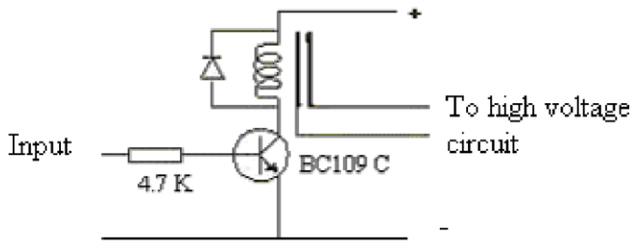


Figure 4.6

Use BC109C or a similar type as the transistor. The value of resistor will be about 4.7 kΩ. The diode is needed to short circuit the high voltage "back emf" induced when the current flowing through the coil is suddenly switched off.

Exercise 03

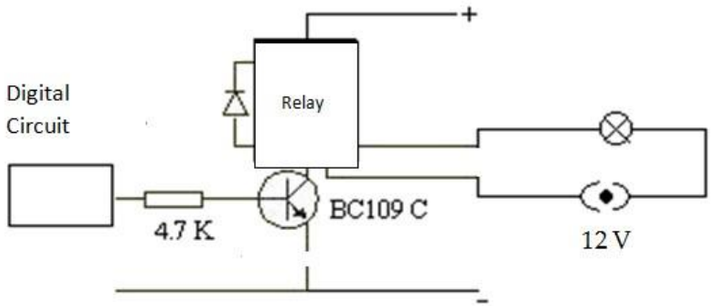


Figure 4.7

1. Construct the figure 4.7 circuit and test whether the bulb functions according to the conditions mentioned in exercise 02.
2. Discuss the function of the transistor used in this circuit.
3. What will happen if the diode is removed from the circuit?

Exercise 04

1. Think of a problem.
2. Design the truth table for the above mentioned problem.
3. Write down a logic expression.
4. Draw a circuit diagram using basic logic gates.
5. Construct the circuit and check it.

Experiment No: 05

Deflection Magnetometer Experiment

5.1. Investigation of Inverse Square Law by Gauss' Method

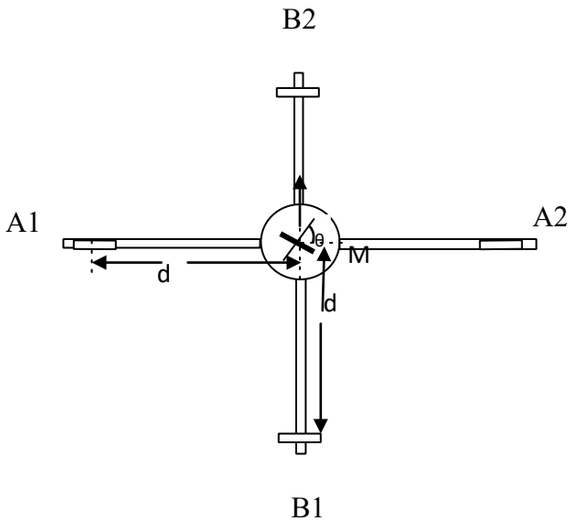


Figure 5.1

Theory:

According to an inverse n^{th} power law;

Magnetic field strength at position A,

$$H_A = \frac{nM}{d^{n+1}} \times \frac{\mu_0}{4\pi}$$

At position B,

$$H_B = \frac{M}{d^{n+1}} \times \frac{\mu_0}{4\pi}$$

Where, M – Magnetic moment of the magnet

Then,

A Position

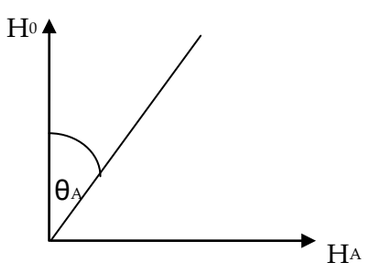


Figure 5.2

$$Tan\ \theta_A = \frac{H_A}{H_0}$$

B Position

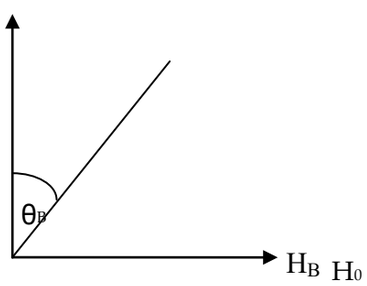


Figure 5.3

$$Tan\ \theta_B = \frac{H_B}{H_o}$$

$$\frac{Tan\ \theta_A}{Tan\ \theta_B} = \frac{H_A}{H_B} = n$$

Apparatus

Deflection magnetometer, meter or half meter rulers, a short strong bar magnet.

Exercise 01

1. Arrange the deflection magnetometer M with its pointer at zero, and with rules below it in both the north-south and east-west directions.
2. Place the short magnet B_1 in the B (broadside) position so that: (a) the distance d of its mid-point from M is at least 5 times its half-length; (b) the deflection is between 30^0 and 60^0. Note d and the reading of both ends of the pointer, tapping the magnetometer gently to reduce the effect of friction on the needle.
3. Reverse B_1 so that its poles are interchanged and again read both ends of the pointer. Then take B_1 to B_2 at exactly the same distance d on the other side of M and repeat the four readings.
4. Now place the magnet in the A (end – on) position at A_1, so that the distance d as before. Take the two readings of the pointer, reverse the poles at A_1 and again take the readings. And then place the magnet at A_2 on the other side of M and take the four readings again.
5. Repeat for four other distances d, keeping d at least 5 times the half – length of the magnet and the angle between 30^0 and 60^0.

Graph

When you draw $\frac{Tan\ \theta_A}{Tan\ \theta_B}$ ***Vs d*** it should behave as the figure 5.4 and show the value of n tends to 2 when d is large compared with l, the magnetic half – length of the magnet.

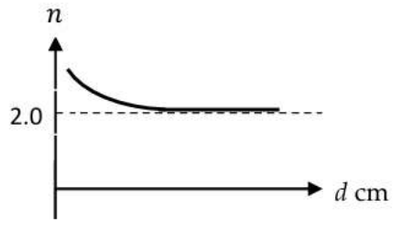


Figure 5.4

Measurements

Distance d (cm)	Deflections in Gauss B Positive (θ_B)	Deflections in Gauss A Positive (θ_A)	_____
	Average =	Average =	

5.2. Measurement of Magnetic Length by Deflection Magnetometer

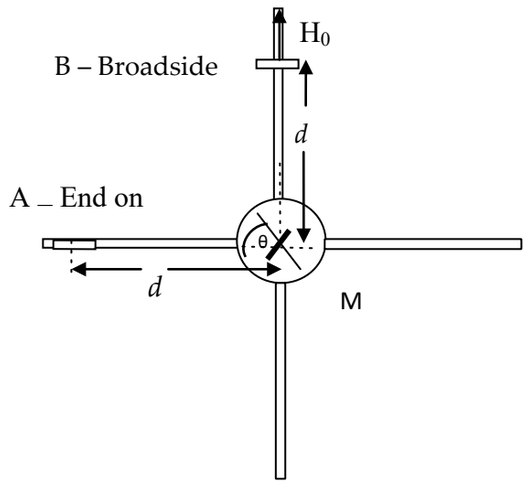


Figure 5.5

1. Broadside (B) Position Theory:

With the usual notation.

Where l – half length of the magnet, M – magnetic moment

$$H = \frac{\mu_0}{4\pi}\frac{M}{(d^2+l^2)^{3/2}} = H_0 tan\theta$$

$$(d^2+l^2)^{3/2} = \frac{\mu_0 M}{4\pi H_0}\cot\theta$$

$$d^2+l^2 = \left(\frac{\mu_0 M}{4\pi H_0}\right)^{2/3}(\cot\theta)^{2/3}$$

$$(\cot\theta)^{2/3} = \left(\frac{4\pi H_0}{\mu_0 M}\right)^{2/3} d^2 + \left(\frac{4\pi H_0}{\mu_0 M}\right)^{2/3} l^2$$

Graph

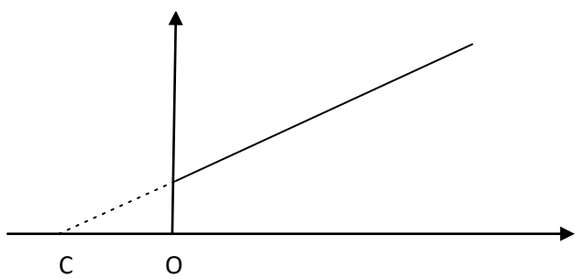


Figure 5.6

Plot a graph of $(\cot\theta)^{2/3}$ **Vs** d^2. Draw the best straight line through the points and extrapolate it back to C on the d^2 – axis.

When $(\cot\theta)^{2/3} = 0$, then $d^2 + l^2 = 0$ or $d^2 = -l^2$

$l^2 = OC$; $l = \sqrt{OC}$

Magnetic length = $2l = 2\sqrt{OC}$

Apparatus

Deflection magnetometer, meter or half meter ruler, long bar magnet.

Exercise 02

1. Arrange the magnetometer (M) so that the pointer reads zero. Then place the magnet at B in the broadside position (i.e. pointing east-west) and move it until the deflection is between 30^0 and 60^0.
2. Take the readings at both ends of the pointer, reverse the magnet poles and take the readings again, and then place the magnet at exactly the same distance on the other side and again take four readings of the deflection.
3. Repeat the eight readings for four more broadside distances, d from M the deflection being between 30^0 and 60^0.

Measurements

Distance d (m)	Deflections	Average Deflection (θ)	d_2	$(\cot\theta)^{2/3}$

Experiment No: 6

Comparison of Magnetic Moments / Deflecting Magnetometer

6.1. Comparison of Magnetic Moment of two Bar Magnets using the Deflection Magnetometer

Theory:

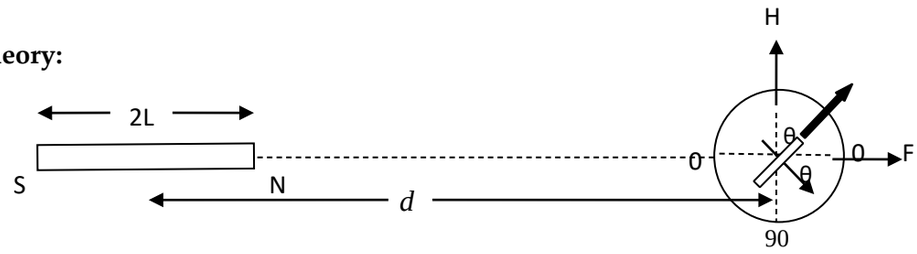


Figure 6.1

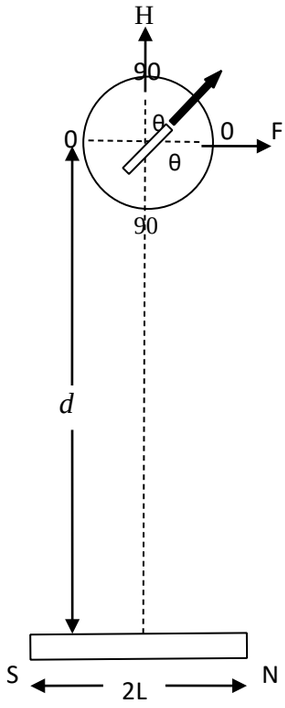


Figure 6.2

The magnetic field intensity F along the axis of a bar magnetic momentum "M" and the magnetic length 2*l* is given by;

$$F = \frac{\mu_0}{4\pi}\frac{2Md}{(d^2 - l^2)^2}$$

Where, d - the distance from middle of the bar magnet to the point of interest.

If the magnetic axis of the magnet is normal to the earth's magnetic polar line at a location (Figure 6.1), then the deflection of magnetometer kept at that point will be given by following expression,

$$F = \frac{\mu_0}{4\pi}\frac{2Md}{(d^2 - l^2)^2} = H\tan\theta$$

Where, H – the horizontal component of the earth's magnetic field.

$$\frac{M}{H} = \frac{4\pi}{\mu_0}\frac{(d^2 - l^2)^2}{2d}\tan\theta$$

Exercise 01

1. First keep the bar magnet away and rotate the magnetometer until the indicator aligns the 0-0 diameter.
2. Keep the meter ruler on the magnetometer along the indicator of the meter. Now the meter ruler is in the direction of magnetic East-West direction.
3. Place the magnet on a piece of wood and keep it under the meter ruler.
4. Set it at a sufficient length (d_1) from the magnetometer while its axis is exactly under the numbered edge of the ruler. Write down the two readings (θ_1, θ_2) of the magnetometer.

5. Now keep the magnet at the same position and interchange its North and South poles, take the new reading (θ_3, θ_4). Repeat same measurements for two more different "d" values in the same side of the magnetometer while keeping the deflection between 30^0 and 60^0.
6. Repeat the same measurements at the opposite side of the magnetometer and take the corresponding values θ_5, θ_6, θ_7, θ_8 for the above mentioned "d" values.
7. Note down the geometrical length of the magnet and assume that the magnetic length is 5/6 of the geometrical length.
8. Take the average value of eight deflections of each distance and calculate the value M_1/H.
9. Take the average value of these values as the final M_1/H.
10. Similarly use this method to find M_2/H for the other bar magnet.
11. The ratio of these two values is the ratio of magnetic moments.

When the magnet is placed as in the figure 6.2, the field intensity at the point on the equator of a bar magnet;

$$F = \frac{\mu_0}{4\pi}\frac{M}{(d^2+l^2)^{3/2}} = H\tan\theta$$

$$\frac{M}{H} = \frac{4\pi}{\mu_0}(d^2+l^2)^{3/2}\tan\theta$$

Repeat the experiment while placing the magnet as figure 6.2.

6.2. Comparison of Magnetic Moment of two Bar Magnets using Oscillating Magnetometer.

Theory:

When a magnet with magnetic moment M oscillates in the field of H, the period T is;

$$T = 2\pi\sqrt{\frac{I}{MH}}$$

Where, I = Moment of inertia of the magnet about the oscillating axis.
i.e.

$$I = \frac{\omega(a^2 + b^2)}{12}$$

a, b – Length and the width of the magnet
– Mass of the magnet (Use the chemical balance)

Now for the two magnets;

$$\frac{M_1}{M_2} = \frac{I_1}{I_2}\left(\frac{T_2}{T_1}\right)^2$$

Exercise 02

1. Mark the magnetic meridian on the table with the compass.
2. Now place a brass bar on the stand when it is at rest, adjust both brass bar and the line in the magnetometer box until they become parallel to the magnetic meridian.
3. Now replace the brass bar with the bar magnet with its North Pole in the direction of North.
4. Now oscillate the magnet and measure the time period for twenty five complete oscillations. Hence find the period ($_1$). Similarly, find the period ($_2$) for the other bar magnet.
5. Now measure the lengths, widths and mass of each magnet. Using the above expression, obtain the ratio of magnetic moments of the two bar magnets.

6.3. Finding the Neutral Point of a Magnetic Field by the Track Method and Calculating the Magnetic Moment of a Magnet by this Method

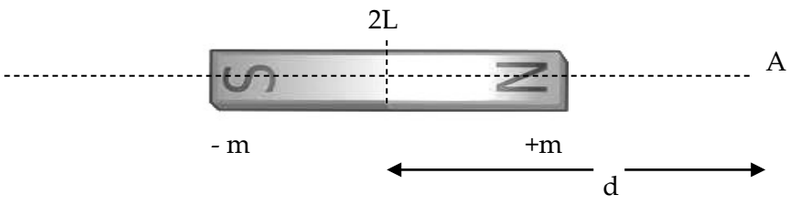


Figure 6.3

If A is a point on the axis, at the distance d from the middle of a bar magnet which has the pole strength "m" and the magnetic length "2L";

Then the field intensity at A;

$$F_A = \frac{\mu_0}{4\pi}\frac{2MD}{(d^2 - l^2)^2}$$

Where, M is the magnetic moment (2m)

F_A and H must be equal and opposite to become the point A neutral, where the horizontal component of the earth's magnetic field is H.

Then,

$$F_A = H = \frac{\mu_0}{4\pi}\frac{2MD}{(d^2 - l^2)^2}$$

Now consider a point (B) on the magnetic equator (i.e. Normal to axis of the magnet and also going through its midpoint)

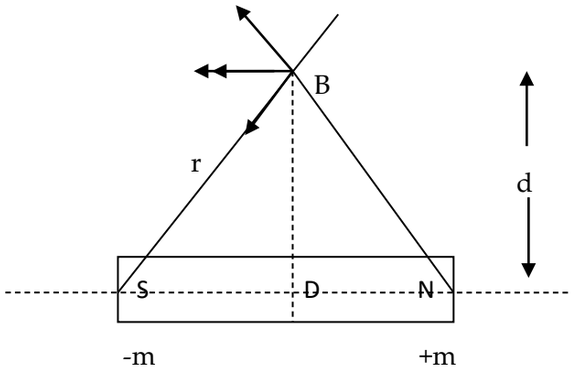


Figure 6.4

Similarly the field at the point B;

$$F_B = \frac{\mu_0}{4\pi}\frac{2ml}{r^2}cos\theta = \frac{\mu_0}{4\pi}\frac{M}{(d^2 + l^2)^{3/2}}$$

In the direction of NS. Hence for B to become a neutral point;

$$H = \frac{\mu_0}{4\pi}\frac{M}{(d^2 + l^2)^{3/2}}$$

Exercise 03

1. Fix a graph paper on the drawing board as its lines are parallel or normal to the magnetic meridian. Place the given bar magnet in the way that its North Pole directs towards the South and parallel to the magnetic meridian. After that mark its outline on the paper.
2. Roughly find whether the Null point (Neutral Point) is on the paper. Place the compass on the graph paper and obtain the points where its indicator is normal to the magnetic meridian.
3. Draw a smooth curve by connecting these points. Also obtain the points where the indicator of the compass parallel to the magnetic meridian and connect them.
4. Obtain the points where the above mentioned curves interchange. Find the places where the magnetic poles exist by drawing magnetic field lines near the poles.
5. Repeat the experiment in the way that the North Pole of the magnet directs towards the North.
6. Take H = 4×10^{-5} T and calculate the magnetic moments at the above mentioned two cases and take the average of them.

Permeability of free space, $_0 = 4\pi \times 10^{-7}\ Hm^{-1}$

Complete all the error calculations using the error theory propagation and draw error bars for the x and y data points if possible before you draw the graph for all the relevant exercises.

Experiment No: 07

Voltage Regulation

Part (A). Studying Oscilloscope and Signal Generator

Introduction to apparatus

Oscilloscope:

- Analogue Cathode Ray Oscilloscope - Hameg HM 400
- 2 Channels
- Max Sensitivity 1mV div

Signal Generator

- 100 kHz

Exercise 01

1. Switch the oscilloscope.
2. Observe the screen. You can see one or two tracing lines on the screen
3. Observe the entire screen appearance and note the vertical and horizontal grids on the screen.
4. Press GND button/switch (**25**) and set the Dual mode (**32**). Now the two horizontal tracing lines can be seen.
5. By adjusting INTENSITY (push button (**3**)) observe the effects on tracing lines and state your observations.
6. Repeat the step 5 for the FOCUS control (**3**) also.
7. Change the setting of the following control knobs and note down your observations.
   X-POSITION (**13**)
   POSITION 1 (**5**)
   POSITION 2 (**5**)
8. Plug the test probe to CH2 BNC connector (**29**).
9. Connect the signal input probe to the 1 KHz/1 MHz square wave output (**30**) while connecting crocodile clip of the probe to the ground connection (**30**). (It is not necessary to connect the ground of your test probe for calibration).

10. Press the button for DC position (**24**). Make sure the GND button depressed. Turn the VOLTS/DIV (V) (**14**) rotary switch to 0.1V position and TIME/DIV (t) (**15**) rotary switch to 0.5 ms position.
11. By adjusting the control knobs of X – POSITION and POSITION 2, take the following readings.
    a) Number of vertical grids (Y) from GND line position to peak of the wave form
    b) Number of horizontal grids (X) for the period of the wave form
    c) Reading of VOLTS/DIV rotary switch
    d) Reading the TIME/DIV rotary switch

    Amplitude and periodic time of the wave can be calculated as,
    Amplitude = Y.V
    Periodic Time = X.t

12. Fill the table given below

	CH 2
Number of vertical grids	
Number of horizontal grids	
VOLTS/DIV reading	
TIME/DIV	
Amplitude	
Periodic Time	
Wave form	

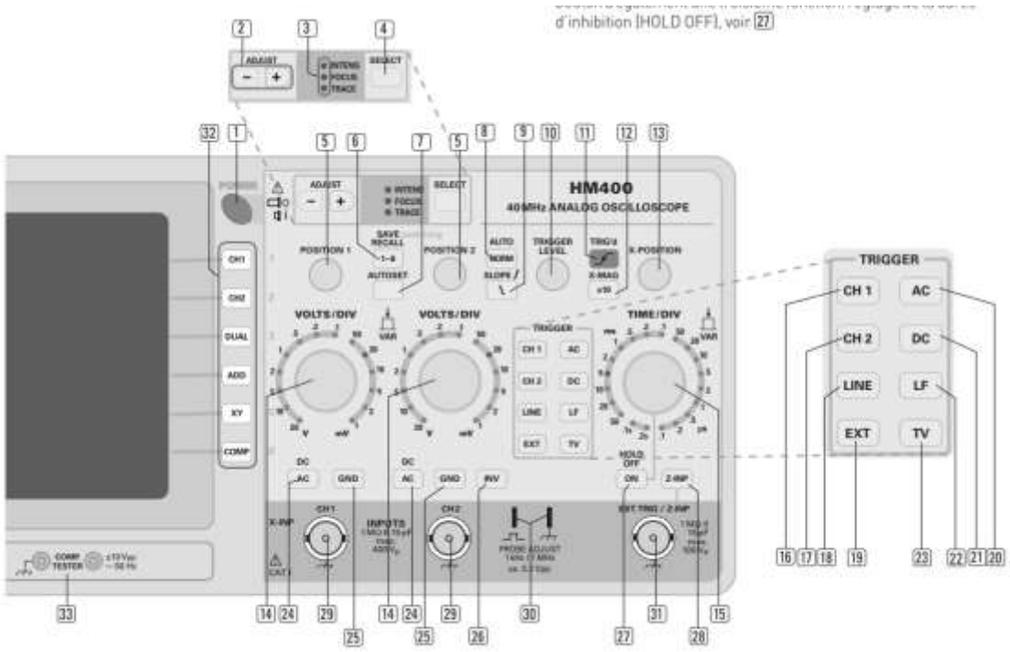


Figure 1: Controls of Analog Cathode Ray Oscilloscope

13. Press the INV push button and draw the waveform. Fill the table below.

	CH 2
Wave form when switch depressed	
Wave form when switch pressed	

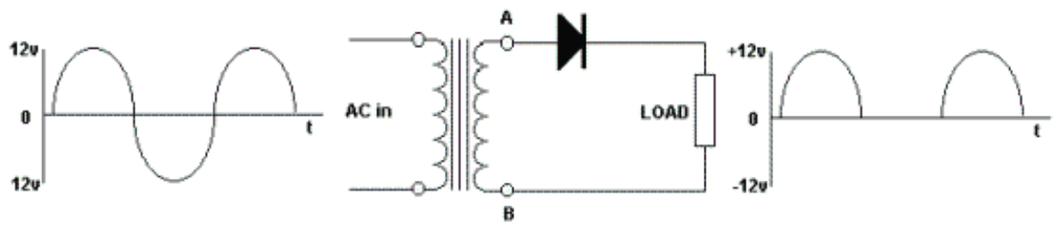
14. Set the push button to the AC position and note the effects. Draw the waveforms for the each case in the given table.

	GND	**AC**	**DC**
CH 2			

Part (B). Wave Rectification

Wave rectification, smoothing and regulation circuits are mainly use in DC power supply units and their main steps can be recognized as follows. The first step of a power supply is rectification. There are two types of rectifiers as half wave rectifiers and full wave rectifiers.

(a). Half wave Rectification



The voltage at point A is constantly changing its polarity between positive and negative. This is known as an AC waveform.

During the first half cycle of the waveform shown on the left, A is positive. The diode is forward biased and current flows around the circuit formed by the diode, the transformer winding and the load.
Since the current flows through the load, and the voltage across the load is proportional to it, then the voltage across the load is as shown in the right hand diagram, during the first half cycle.

During the second half cycle, B (cathode end) is positive. The diode is reverse biased and no current flows. This is indicated by the horizontal line in the right hand diagram. The diode only conducts on every other half cycle.

Exercise – 02 (Half Wave Rectifier)

(i) Test your transformer: give 230V, 50Hz source to the primary coil of the transformer and observe the AC waveform of rated value without any distortion at the secondary of the transformer (in CRO CH1-A and CH2-B).

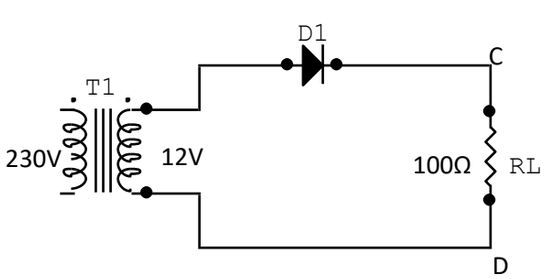
(ii) Connect your circuit to the secondary terminals of the transformer and connect CRO across the load (CD). Record the waveform.

(iii) Connect the circuit with a capacitor (1000 μF) as shown in following figure. Record the waveform.

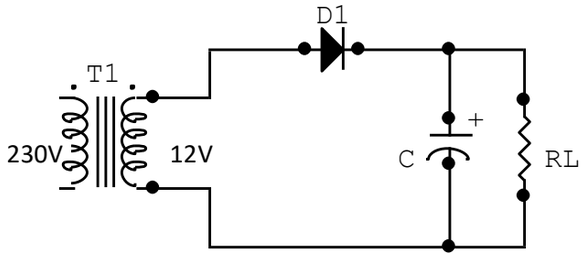
(iv) While keeping the C, change the R_L and measure the V_{DC} for each R_L by using CRO. (Note down the output signal form, and ripple voltage, R_L =330 Ω, 1 kΩ, 3.3 kΩ, 4.3 kΩ and 10 kΩ).

(v) Calculate the I_{DC} and plot the Voltage Regulation Curve (V_{DC} Vs I_{DC})

(vi) While keeping R_L=10 kΩ, change the C and observe the output through CRO. (Note down the output signal form and ripple voltage, C = 680 μF, 1000 μF and 2200 μF).



Step 1



Step 2

(b). Full wave Rectification

Full wave rectification can be conducted using two methods

(i) Full wave rectification using two diodes and a center-tap transformer

(ii) Full wave rectification using four diodes (diode bridge)

Exercise – 03 (Full Wave Rectifier Using Center tap Transformer)

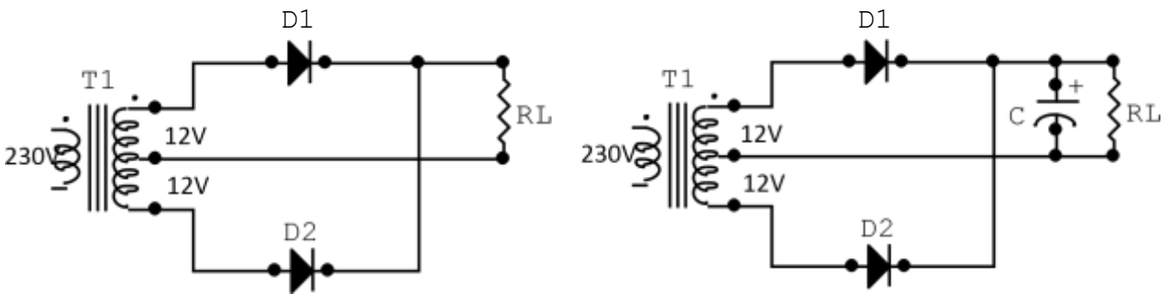
(i) Connect your circuit to the secondary terminals of the transformer and connect CRO across the load.

(ii) Connect the circuit with a capacitor (1000 μF) as shown in following figure.

(iii) While keeping the C, change the R_L and measure the V_{DC} for each R_L by using CRO. (Note down the output signal form, and ripple voltage, R_L = 330 Ω, 1 kΩ, 3.3 kΩ, 4.3 kΩ and 10 kΩ).

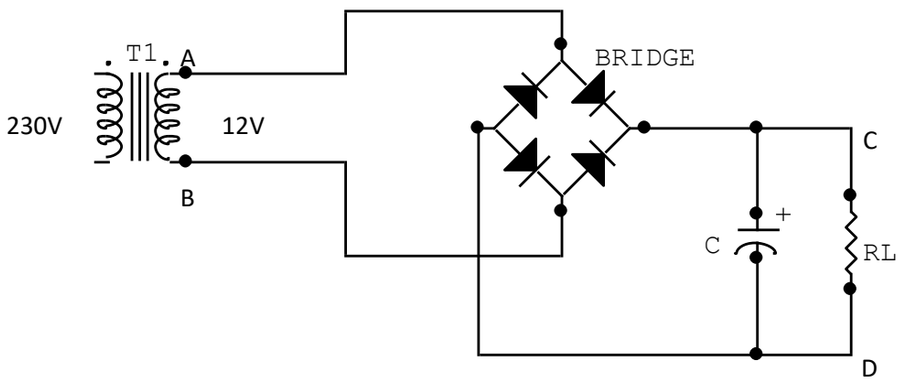
(iv) Calculate the I_{DC} and plot the Voltage Regulation Curve (V_{DC} Vs I_{DC})

(v) While keeping R_L=10 kΩ, change the C and observe the output through CRO. (Note down the output signal form and ripple voltage, C = 100 μF, 220 μF, 470 μF, 680 μF and 1000 μF).



Exercise – 04 (Full Wave Rectifier Using a Diode Bridge)

(i) Construct the following circuit and connect the CRO across the load



(ii) Change the capacitor value from 100 μF to 1000 μF while keeping the value of R_L at 3.3 kΩ. Observe and record the shape of the waveform across the AB, CD and measure V_{DC}.

Resistor Value	Capacitor value	Ripple Voltage	Waveform
3.3 kΩ	100 μF 220 μF 1000 μF		

Experiment No: 08

Operational Amplifier (OP-AMP)

Introduction:

The op-amp is basically a differential amplifier having a large voltage gain, very high input impedance and low output impedance. The op-amp has an "inverting" or (-) input and "non-inverting" or (+) input and a single output.

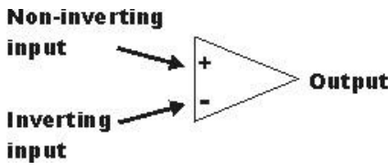


Figure 8.1

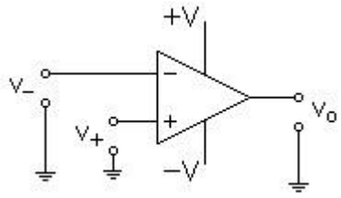
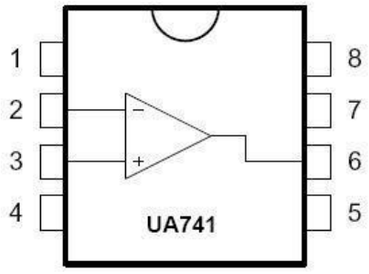


Figure 8.2

The op-amp is usually powered by a dual polarity power supply in the range of -15 volts to +15 volts for UA741 and it is supplied in an 8 pin DIP (Dual Inline Package)





1 - Offset null 1
2 - Inverting input
3 - Non-inverting input
4 - V_{CC}^-
5 - Offset null 2
6 - Output
7 - V_{CC}^+
8 - N.C.

Figure 8.3

8.1 Non-inverting Amplifier

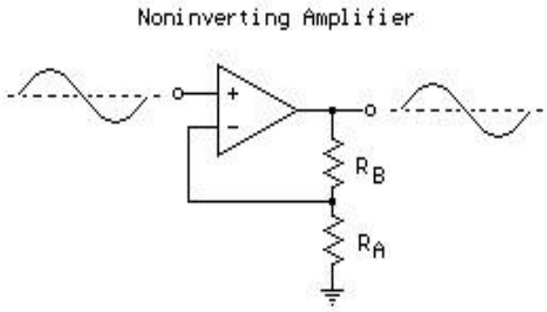


Figure 8.4

The non-inverting amplifier is connected so that the input signal goes directly to the noninverting input (+) and the input resistor R_A is grounded. In this configuration, the input impedance as seen by the signal is much greater since the input will be following the applied signal and not held constant by the feedback current. As the signal moves in either direction, the output will follow in phase to maintain the inverting input at the same voltage as the input (+). The voltage gain is always more than one and is,

$$A_V = V_{gain} = \frac{V_{out}}{V_{in}} = \frac{R_A + R_B}{R_A}$$

Apparatus

Bread board, 741 IC, Dual power supply, Potential divider, R_A=1kΩ, R_B=1kΩ, Digital multimeter

Exercise 01

1. Construct a non-inverting amplifier using 741 IC with R_A=1kΩ and R_B=1kΩ. Use power supply to provide ±12V to the circuit.
2. Use a potential divider for providing a voltage between -5V to +5V to the input. Vary the input voltage in small steps and measure both input and output voltages using digital voltmeter.
3. Plot a graph of V_{out} *vs* V_{in}.
4. Determine voltage gain and compare with theoretical value.

Complete all the error calculations using the Statistical Method (Least Squared Method).

8.2 Inverting Amplifier

Theory:

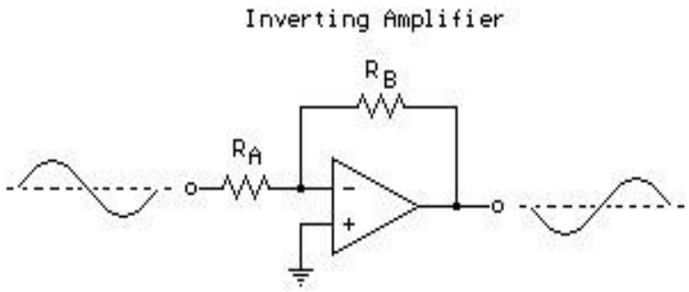


Figure 8.4

Figure 8.5

The op-amp is connected using two resistors R_A and R_B such that the input signal is applied in series with R_A and the output is connected back to the inverting input through R_B. The noninverting input is connected to the ground reference or the center tap of the dual polarity power supply.

During operation, as the input signal goes positive, the output will go negative and vice versa. The amount of voltage changes at the output relative to the input depends on the ratio of the two resistors R_A and R_B.

As the input moves in one direction, the output will move in the opposite direction, so that the voltage at the inverting input remains constant or zero volts.
If R_A is 1K and R_B is 10K and the input is +1 volt then there will be 1 mA of current flowing through R_A and the output will have to move to -10 volts to supply the same current through R_B and keep the voltage at the inverting input at zero. The voltage gain in this case would be,

$$A_V = V_{gain} = \frac{-V_{OUT}}{V_{IN}} = \frac{-R_B}{R_A} = \frac{-10k\Omega}{1k\Omega} = -10$$

Apparatus

Bread board, 741 IC, Dual power supply, Potential divider, R_A=1kΩ, R_B=10kΩ, Digital multi-meter

Exercise 02

1. Construct an inverting amplifier using 741 IC with R_A=1kΩ and R_B=10kΩ. Use power supply to provide ±12V to the circuit.
2. Use a potential divider for providing a voltage between -5V to +5V to the input. Vary input voltage in small steps and measure both input and output voltages using digital voltmeter.
3. Plot a graph of V_{out} vs V_{in}. Determine voltage gain and compare with theoretical value.

Complete all the error calculations using the Statistical Method (Least Squared Method).

8.3. The Comparator (741 Operational Amplifier as a comparator)

Theory:

In here the 741 is used as a comparator and not an amplifier. A comparator is circuit that compares an input voltage with a reference voltage. The output of the comparator then indicates whether the input signal is either above or below the reference voltage (V_{ref}). When the V_{in} is greater than the V_{ref} the comparator output goes to a high level.

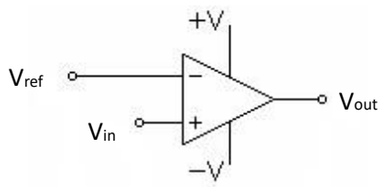
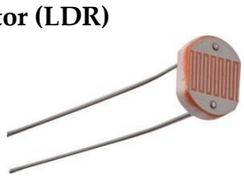


Figure 8.6

Light dependent resistor (LDR)



A photo resistor or LDR is an electronic component whose resistance decreases with increasing incident light intensity. It can also be referred to as a light dependent resistor (LDR).

A photo resistor is made of a high resistance semiconductor. If intensity of light falling on the device is of high enough frequency photons absorbed by the semiconductor give bound electrons enough energy to jump into the conduction band. The resulting free electron (and its whole partners) conduct electricity, thereby lowering resistance.

Dark Activated Lamp

In here 741 IC use as a comparator. From R_1 and R_2 voltage dividers have a constant voltage at point A. That is reference voltage (V_{ref}) of the comparator. Resistance of the LDR can vary according to intensity of the light. Therefore, voltage at point B (V_{in}) also can vary.

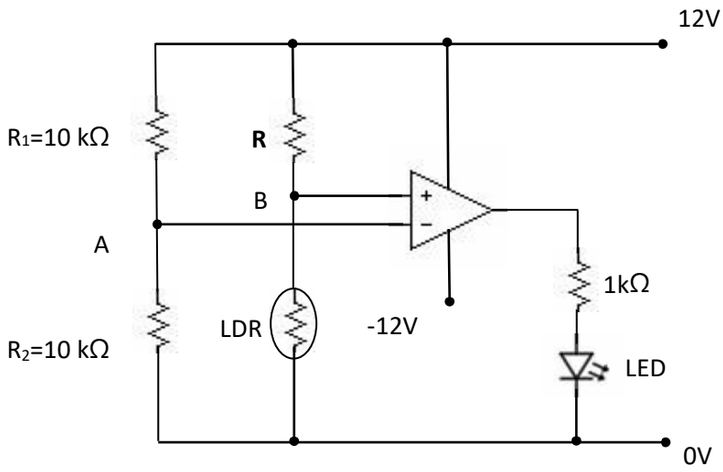


Figure 8.7

Exercise 03

1. Measure the resistance of LDR in dark and light conditions.
2. Choose a suitable value for **R** in the circuit.
3. Construct the above circuit and measure the voltage at point A and point B in dark and illuminated states. Explain how it works as a comparator.
4. Change this circuit to construct a light activated lamp.

8.4 Inverting Summing Amplifier

Theory:

It is very common in circuit design to combine several signals into a single common signal. One good example of this is in the broadcast and recording industries. The typical modern music recording will require the use of perhaps dozens of microphones, yet the final product consists of only two stereo output signals. If signals are joined haphazardly, excessive interference, noise & distortion may result. The ideal summing amplifier would present each input with an isolated load not affected by other channels.

The most common form of summing amplifier is really nothing more than an extension of the inverting voltage amplifier. The simple summing amplifier shown in Figure 8.8 is capable of summing as many signals as desired with a proportionality determined by the ratio of feedback resistance to input resistance.

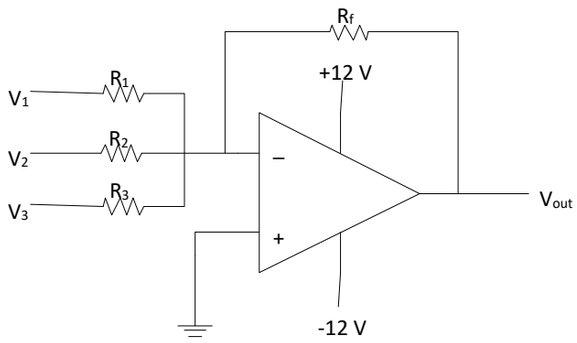


Figure 8.8

By analyzing the figure, we can write an equation for output in terms of inputs as follows,

$$V_{out} = -\left(\frac{R_f}{R_1}V_1 + \frac{R_f}{R_2}V_2 + \frac{R_f}{R_3}V_3\right)$$

Apparatus

Bread board, 741 IC, Dual power supply, Potential divider, $R_1 = R_2 = R_3 = 1\ k\Omega$, $R_f = 1\ k\Omega$, Digital multi-meter

Exercise 04

1. Construct an inverting summing amplifier (Figure 8.8) using 741 IC with $R_1 = R_2 = R_3 = 1\ k\Omega$ and $R_f = 1\ k\Omega$. Use power supply to provide ±12V to the circuit.
2. Use a potential divider for providing a voltage between -5V to +5V to the inputs. Vary the input voltages in small steps and measure both inputs and output voltages using digital voltmeter.
3. Fill out the given table and discuss the functionality of the circuit.
4. Discuss how you use this circuit to design a digital to analog converter.

V_1	V_2	V_3	V_{out}
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	
.	.	.	
.	.	.	
.	.	.	

Experiment No: 09

Bifilar Suspension

9.1 Investigation of the factors determining the period of oscillation of a Bifilar Suspension.

Theory:

The bifilar suspension is a technique used to determine the moment of inertia of any type of object about any point on the object. This is done by suspending two parallel cords of equal length through the object examined.

This experiment's main objective is to determine the moment of inertia of a rod by suspending the cords through the mass center of bodies, obtaining an angular displacement about the vertical axis through the center of mass by a sensibly small angle.

This experiment can be used to investigate how the period of oscillation varies with the distance d between the vertical suspension threads (keeping l constant) or distance l, the suspension length (keeping distance d constant). The suspension may also be used to determine the radius of gyration of any body. In this case the body under investigation is bolted to the center of the suspended bar and the periodic time of the combined bar and body measured.

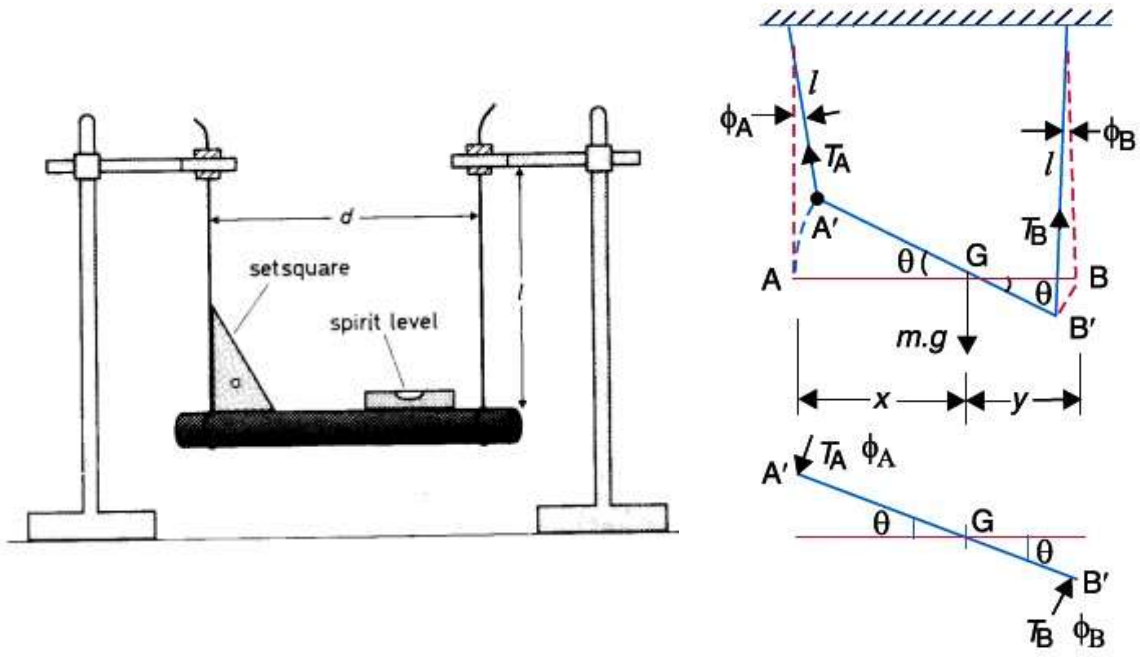


Figure 9.1: Bifilar pendulum apparatus.

- The body whose moment of inertia is to be determined (say AB) is suspended by two long parallel flexible strings as shown in Fig 9.1. When the body is twisted through a small angle θ about a vertical axis through the center of gravity *G*, it will vibrate with simple harmonic motion in a horizontal plane.

m = the mass of the body,

I = the moment of inertia of the body about a vertical axis pass through the center of the rod,

l = the length of each string, (always equal for the first task)

x = the distance between A and G (i.e. AG),

y = the distance between B and G (i.e. BG),

θ = Small angular displacement of the body from the equilibrium position in the horizontal plane,

φ_A and φ_B = the corresponding angular displacements of the strings,

When the body is stationary, the tension in the strings are given by taking moments about B and A respectively,

$$T_A = \frac{m.g.y}{x+y}, \quad \text{and} \quad T_B = \frac{m.g.x}{x+y}$$

- When the body is displaced from its equilibrium position in a horizontal plane through a small angle θ, then the angular displacements of the strings are given by Φ_A and Φ_B,

$$\therefore \quad \phi_A = \frac{x.\theta}{l}; \qquad \text{and} \qquad \phi_B = \frac{y.\theta}{l}$$

- These components of tensions T_A and T_B are equal and opposite in direction, which gives rise to a couple.

- The couple or torque applied to each string to restore the body to its initial equilibrium position, i.e. restoring torque.

$$= T_A.\phi_A.x + T_B.\phi_B.y$$

$$= \frac{m.g.x.y.\theta}{l\,(x+y)}\,(x+y)$$

$$= \frac{m.g.x.y.\theta}{l}$$

The equation of the angular motion is given by $\frac{Id^2\theta}{dt^2} = -\frac{mgd^2}{4l}\theta$

$$\frac{d^2\theta}{dt^2} + \frac{mgd^2}{I*4l}\theta = 0$$

$$\ddot{\theta} + \omega^2\theta = 0$$

From the above last equation,

$\omega = \sqrt{\frac{mgd^2}{I*4l}}$ Where ω is the angular velocity of the bar.

Also $T = \frac{2\pi}{\omega}$ where T is the periodic time.

Therefore;

$$\boldsymbol{T = 4\pi\sqrt{\frac{Il}{mgd^2}}}$$

Apparatus

A metal rod, meter ruler, strings, stop watch, triple balance and lab stands.

Procedure 9.1To investigate how the period of oscillation varies with the length *l* of the suspension threads

- Set up the equipment as shown in figure 9.2.
- Adjust the spacing of the strings so that they are 60 cm (distance between one end of the rod and string is 20 cm).
- Adjust the length of the strings to 80 cm.
- Hold the body and rotate it around the center (by a small angle around 15^0).
- Measure the time taken for a known amount of oscillations. (about 10)
- Repeat the above measurement for about five times.
- Repeat the above process for different *l* lengths of the string.

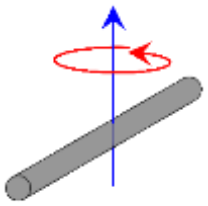


Figure 9.2: A metal rod rotating about the center of the plane surface

Exercise 01

1. In experiment 1, plot a correct graph selecting the correct axes with the values T and l.
2. Measure the slope and interpret your result of the graph.
3. Find the moment of inertia of the rod system using your results.
4. **Complete all the error calculations using the error propagation formula.**
5. Calculate the uncertainty (error) of the moment of Inertia I.

Constant distance apart of vertical threads d=									
Length of vertical threads l	Time for 10 oscillations						Time for 1 oscillation T /s	Log T	Log l
	T_1	T_2	T_3	T_4	T_5	Mean T			

9.2. To investigate how the period of oscillation varies with the distance d between the vertical threads.

Procedure

1. Set up the equipment as shown in figure 9.1.
2. Adjust the spacing of the strings so that they are 60cm both at the top and bottom of the strings.
3. Adjust the distance between the vertical threads to 80 cm.
4. Hold the body and rotate it around the center (by a small angle around 15^0).
5. Measure the time taken for a known amount of oscillations. (About 10)
6. Repeat the above measurement for about five times.
7. Repeat the above process for different distance d (70 cm, 60 cm, 50 cm and 40 cm) of the string.

Exercise 02

1. In experiment 2, plot a graph by selecting correct dependent and independent variables and choosing the X and Y axes correctly.
2. Measure the slope and interpret your result of the graph.
3. Calculate the moment of inertia *I* of the rod system.
4. Calculate the uncertainty of *I.*
5. Use both methods to find the average value of the moment of Inertia with the uncertainty.
6. Discuss the precautions you have taken to avoid errors in the experiment.

Experiment No: 10

Characteristics of Diodes and Semiconductor properties (Determination of Plank's and Boltzmann constants)

Part (a). Characteristics of Diodes

Introduction

A diode is a specialized electronic component used in many applications. It has two electrodes called the ***Anode*** and the ***Cathode***. Diodes are constructed of semiconductor material, such as silicon, germanium, or selenium.

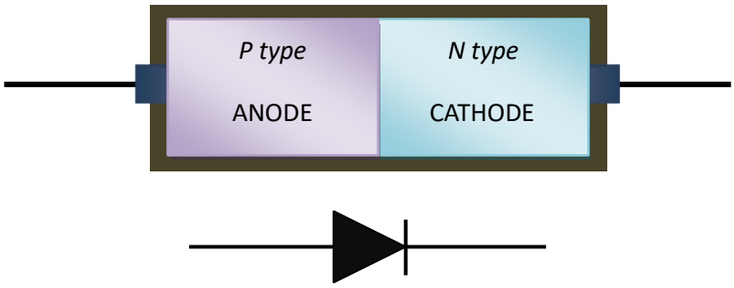


Figure 10.1

Two blocks of N and P type materials are joined together to form a diode and the region where the two materials are joined is called a PN junction. Above figure shows internal construction of a general-purpose diode and its symbol.

Diodes protect circuits from incorrect polarity power connection (switch, signal limiter voltage regulator), convert A.C. into D.C. by blocking one direction of current flow (rectifier), demodulate Amplitude Modulated radio signal (modulators, signal mixers, signal demodulators, and oscillators) etc.

The fundamental property of a diode is its tendency to conduct electric current in only one direction. The current *vs* voltage of an ideal diode behaves as shown in the graph. Forward biased silicon diodes have 0.7 Volts (700mV) across them, in reality.

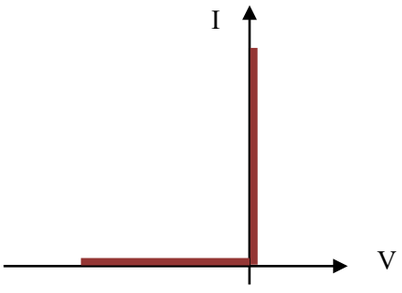


Figure 10.2

The characteristic curve of a real diode is given in the figure 10.3. It has three distinctive regions, the forward bias region, the reverse bias region and the breakdown region.

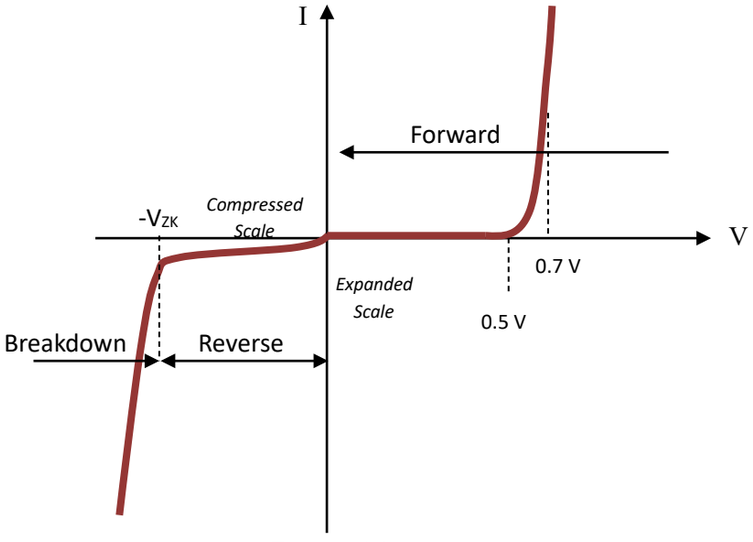


Figure 10.3

The cathode is always identified by a dot, ring or some other mark.

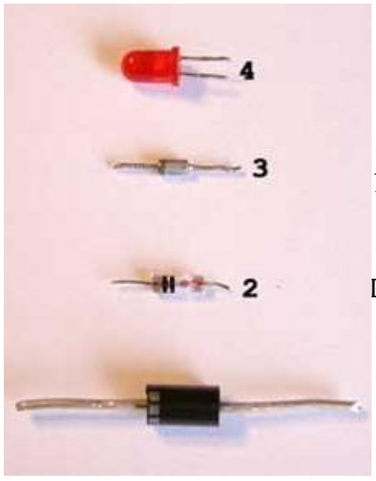


Different type of semiconductor diodes –

There are different types of diodes used for different applications. The LED of the light emitting diode illuminated when it conducts a current. Illuminated LEDs have approximately 2 volts across them. The photodiode is forward biased by incident light and conducts a current when light falls on it. Photodiodes are used as light sensors. The Zener diode is a special type of diode which operates in the reverse breakdown region and provides a constant voltage in the reverse bias mode. Zener diodes are very useful as voltage regulators as they provide accurate reference voltage.



Diode | Light emitting diode (LED) | Photodiode (PD) | Zener diode



Diode

Diode

Power Diode

Figure 10.4

Exercise 01 (First method for testing the diode)

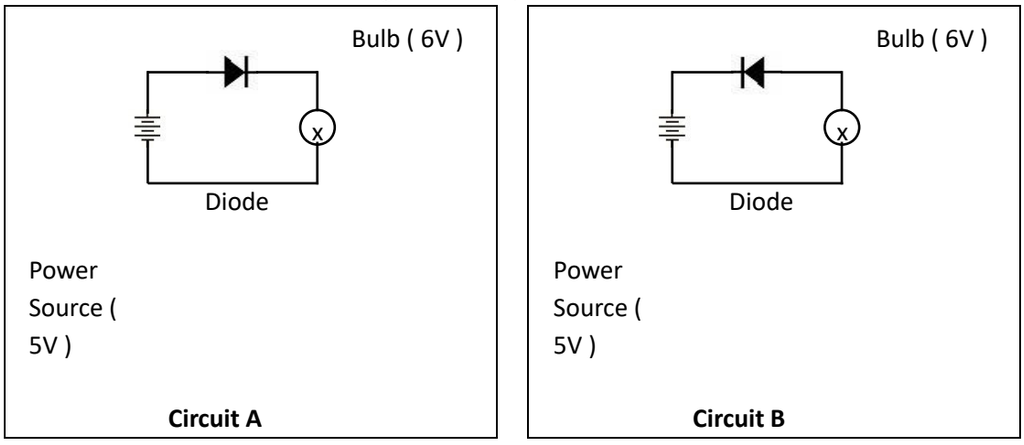


Figure 10.5

(i) Construct circuit **A** shown above.

(ii) Observe whether the bulb is illuminated

(iii) Also construct circuit **B** and test the bulb

(iv) Observe and discuss the difference between circuit **A** and **B**

(Second method for testing the diode)

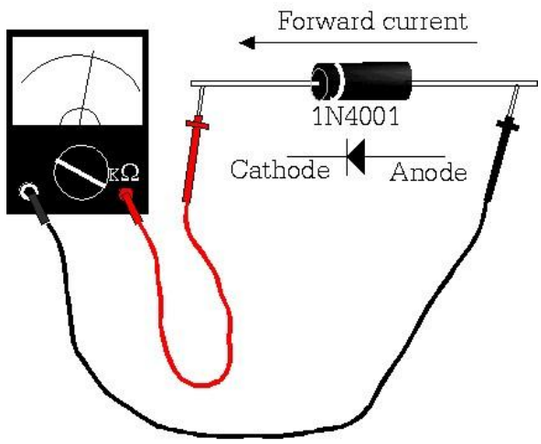


Figure 10.6

(i) Change the analog multimeter scale to kΩ range.
(ii) Place the probes on to the diode terminals as shown above.
(iii) Observe the meter reading while changing the polarity of meter and understand how to test a diode.

Exercise 02 (Diode characteristic curve)

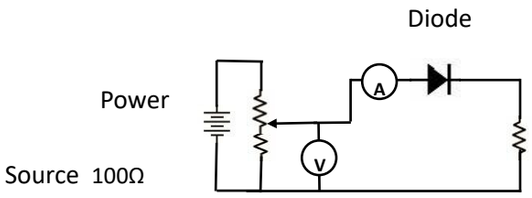


Figure 10.7

(i) Construct the circuit shown above.
(ii) By varying the voltage between 0 to 5V, measure the V and I values.
(iii) Draw the current (I) *vs* voltage (V) graph for a silicon diode. Also discuss the variation of voltage.

Part (B). Determination of Plank's and Boltzmann constants

(a). Determination of Planck's Constant

Theory

Max Planck proposed that atoms absorb and emit radiation in discrete quantities as given by the following equation in order to explain blackbody radiation.

$$E = nhf$$

Where,
n = quantum number
f = frequency of vibration of molecule
h = Planck's constant

Planck's constant can be determined using the characteristics of semiconductors from Modern Physics. Light Emitting Diodes (LEDs) which are semiconductors tend to emit electromagnetic radiation in optical or near optical frequencies when a voltage is applied. LEDs emit light when they are forward biased and above some threshold voltage. These conditions give rise to the concept of electron-hole pairs which are charge carriers in diodes. Electron-hole pairs move under electrical potential thereby produce a current when placed in an electric field. Diodes show exponential behavior above the threshold voltage as shown in figure 10.11. A quantum of energy is being required to create an electron- hole pair, and also equal amount will release when they recombine. In most of the semiconductors or diodes this energy is released as heat. But in LEDs these quanta of energy have been used to produce a photon which gives discrete energy as $E = hf$. Photons with increasing energy will be emitted with the increasing voltage. Also, the light emitted by a LED above the threshold will give discrete wavelengths which are decreasing with increasing energy (voltage).

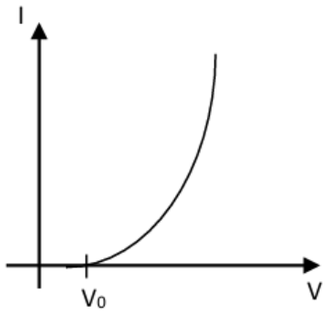


Figure 10. 8

The minimum energy to produce an electron-hole pair which is at maximum wavelength numerically equals to the turn on voltage of the LED. This relation can be expressed by the following equation.

$$E = hf = \frac{hc}{\lambda} = qV_0$$

Where,

c = velocity of light
q = electron charge
λ = wavelength
Other symbols have the usual meanings.

The turn on voltages or the threshold voltages (V_0) which was measured for different colors of LEDs should be considered. Using the gradient of the graph V_0 vs $1/\lambda$ Planck' constant can be determined.

Exercise 03

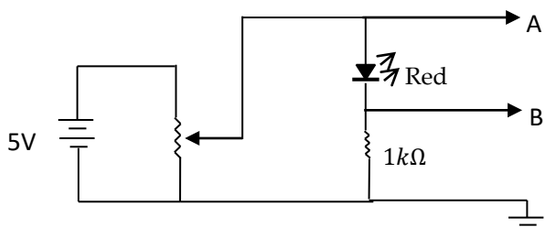


Figure 10.9

1. Construct the circuit shown in Figure 10.9.
2. Obtain the threshold voltage for the LED (Red color) by varying the input voltage.
3. Continue step 2 for the rest of the LEDs in Green, Orange, Yellow, Blue and IR
4. Plot a graph of V_0 vs $1/\lambda$ and determine the Planck's constant using the gradient of the graph.

$c = 3 \times 10^8\, ms^{-1}$
$q = 1.602 \times 10^{-19}\, c$

Color	Red	Green	Orange	Yellow	Blue	IR
Wavelength (nm)	635	570	612	585	470	1000

Complete all the error calculations using the error propagation formula.

(b). Determination of Boltzmann Constant

Theory

The volt-ampere characteristic equation for a PN junction of a semiconductor is given by the Shockley's equation as follows,

$$I = I_s\,(e^{\frac{qv}{kT}} - 1)$$

Where,

I = Diode current
I_s = Reverse leakage current
q = Charge of an electron
v = Voltage of the diode
T = Room temperature in Kelvin
k = Boltzmann constant

When forward bias, the Shockley's equation can be approximated as,

$$I = I_s\,(e^{\frac{qv}{kT}})$$

By considering natural log of the equation,

$$\ln I = \frac{qv}{kT} + \ln I_s$$

Thus the Boltzmann constant can be determined using the gradient of the graph of ln I vs V.

Exercise 04

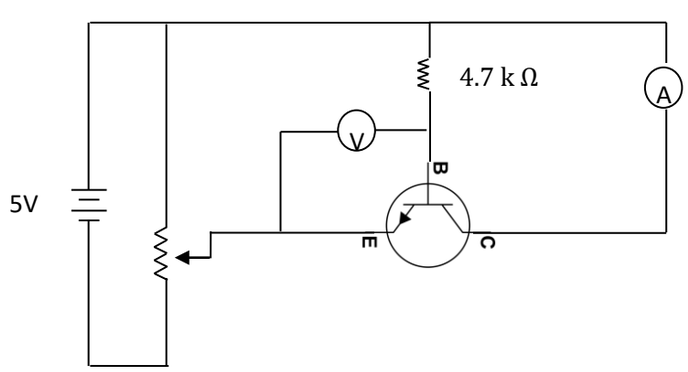


Figure 10.10

1. Construct the circuit shown in Figure 10.10.
2. Measure the Base-Emitter voltage and Collector current by varying the input voltage.
3. Plot a graph of ln I vs V and determine the Boltzmann constant using the gradient of the graph. (Consider room temperature as 300 K)
4. **Complete all the error calculations using the error propagation formula.**